

Problem

The Thevenin impedance of a network seen from the load terminals is $80 + j55 \Omega$. For maximum power transfer, the load impedance must be:

- (a) $-80 + j55 \Omega$
- (b) $-80 - j55 \Omega$
- (c) $80 - j55 \Omega$
- (d) $80 + j55 \Omega$

Step-by-step solution

Step 1 of 3

It is given that, the Thevenin impedance of a network is,

$$\mathbf{Z_{Th} = 80 + j55 \Omega}$$

Write the expression for Thevenin impedance of a network.

$$\mathbf{Z_{Th} = R_{Th} + jX_{Th}}$$

Write the expression for load impedance.

$$\mathbf{Z_L = R_L + jX_L}$$

Step 2 of 3

For the maximum power to be transfer, select $R_L = R_{Th}$ and $X_L = -X_{Th}$. Write the expression for load impedance, $\mathbf{Z_L}$.

$$\begin{aligned}\mathbf{Z_L} &= \mathbf{R_{Th} - jX_{Th}} \\ &= \mathbf{(R_{Th} + jX_{Th})^*} \\ &= \mathbf{Z_{Th}^*}\end{aligned}$$

Therefore, for the maximum power transfer, the load impedance should be equal to the complex conjugate of the Thevenin impedance.

Step 3 of 3

The complex conjugate of the Thevenin impedance is,

$$\begin{aligned}\mathbf{Z_{Th}^*} &= \mathbf{(80 + j55)^*} \\ &= \mathbf{80 - j55 \Omega}\end{aligned}$$

Therefore, the correct option is **(c)**.